Lecture Scribe N-Variate PDF

pdf.pdf
PDF

You are given a lecture PDF used by a professor.

Generate a **lecture scribe** that strictly reconstructs what was taught in class.

Requirements:
- Include ONLY content present in the lecture or directly implied from it.
- Preserve all definitions, theorems, notations, and assumptions exactly.
- Reconstruct all derivations step-by-step, clearly showing how each formula is obtained.
- Include all intermediate steps used in proofs and derivations (no skipping).
- Maintain the same logical flow and dependencies as in the lecture.
- Include worked examples exactly as done in class, step-by-step.
- Do NOT add intuition, simplifications, external explanations, or extra examples.
- Do NOT rewrite as a textbook or tutorial.
- Keep it concise and exam-revision friendly.

Structure:
1. Definitions & Notation
2. Theorems / Statements
3. Derivations (step-by-step, showing origin of formulas)
4. Proofs (complete, no gaps)
5. Worked Examples (as in lecture)
6. Key Results Summary

Output should be clean, structured, and minimal.

Here is the **lecture scribe (Lecture 19: N-Variate Joint PDF/PMF/CDF)** reconstructed strictly from the provided lecture slides .

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1. Definitions & Notation

Joint Gaussian PDF (2 variables)

For random variables X, Y:

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\right.\right.$$

Where:

- μ_X, μ_Y: means
- σ_X, σ_Y: standard deviations

- ρ: correlation coefficient

Correlation Coefficient

$$\rho_{XY} = \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Vector Representation

For $X_1, X_2, \ldots, X_N$:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Joint PMF

$$P_{X_1,\ldots,X_N}(x_1, \ldots, x_N) = P(X_1 = x_1, \ldots, X_N = x_N)$$

Joint CDF

$$F_{X_1,\ldots,X_N}(x_1, \ldots, x_N) = P(X_1 \leq x_1, \ldots, X_N \leq x_N)$$

Joint PDF

$$f_{X_1,\ldots,X_N}(x_1, \ldots, x_N) = \frac{\partial^N}{\partial x_1 \cdots \partial x_N} F_{X_1,\ldots,X_N}(x_1, \ldots, x_N)$$

Independence Condition

If independent:

$$f_X(x_1, \ldots, x_N) = \prod_{i=1}^{N} f_{X_i}(x_i)$$

Marginal PDFs

$$f_X(x) = \int f_{XY}(x, y)\, dy, \quad f_Y(y) = \int f_{XY}(x, y)\, dx$$

Q-function

$$Q(x) = P(Z > x), \quad Z \sim \mathcal{N}(0, 1)$$

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2. Theorems / Statements

- If X, Y are jointly Gaussian:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

- Symmetry of normal CDF:

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$

- Independence criterion:

$$f_{XYZ}(x, y, z) = f_X(x) f_Y(y) f_Z(z)$$

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3. Derivations

(A) Correlation Coefficient

Given:

$$\sigma_X^2 = 1, \quad \sigma_Y^2 = 4, \quad \mathrm{cov}(X, Y) = -1$$

Step 1:

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

Step 2:

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

(B) Joint PDF Construction

Substitute:

$$\mu_X = 2,\ \sigma_X = 1,\ \mu_Y = -3,\ \sigma_Y = 2,\ \rho = -0.5$$

Final:

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right)$$

(C) Marginal PDFs

Using:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

Result:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-2)^2}{2}\right)$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(y+3)^2}{8}\right)$$

(D) Probability Computation

Step 1:

$$P(X < 0) = F_X(0)$$

Step 2 (standardization):

$$Z = \frac{X - \mu_X}{\sigma_X}$$

$$P(X < 0) = P\left(Z < \frac{0-2}{1}\right) = P(Z < -2)$$

Step 3:

$$P(Z < -2) = \Phi(-2)$$

Step 4:

$$\Phi(-2) = P(Z > 2) = Q(2)$$

Final:

$$P(X < 0) = Q(2)$$

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4. Proofs

Independence (Example 2)

Condition:

$$f_{X_1,X_2,X_3}(x_1, x_2, x_3) = f_{X_1}(x_1) f_{X_2}(x_2) f_{X_3}(x_3)$$

LHS:

$$= \frac{1}{8}$$

RHS:

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Thus:

$$\text{LHS} = \text{RHS} \Rightarrow \text{independent}$$

Independence (Example 3)

LHS:

$$f_{XYZ}(x, y, z) = abce^{-(ax+by+cz)}$$

RHS:

$$f_X(x) f_Y(y) f_Z(z) = (ae^{-ax})(be^{-by})(ce^{-cz}) = abce^{-(ax+by+cz)}$$

Thus:

$$\text{LHS} = \text{RHS} \Rightarrow \text{independent}$$

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5. Worked Examples

Example 1: Joint Gaussian

Given:

$$\mu_X = 2,\ \sigma_X^2 = 1, \quad \mu_Y = -3,\ \sigma_Y^2 = 4, \quad \operatorname{cov}(X, Y) = -1$$

Tasks:

- Joint PDF → derived above
- Marginals → Gaussian
- Probability → $Q(2)$

Example 2: Uniform 3D Vector

Region:

$$|x_1|, |x_2|, |x_3| \leq 1$$

(a) Constant

$$\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} C\, dx_1 dx_2 dx_3 = 1$$

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

(b) Marginal f_{X_1,X_2}

$$\int_{-1}^{1} \frac{1}{8} dx_3 = \frac{1}{8}(2) = \frac{1}{4}$$

(c) Marginal f_{X_1}

$$\int_{-1}^{1}\int_{-1}^{1} \frac{1}{8} dx_2 dx_3 = \frac{1}{2}$$

(d) Conditional

$$f_{X_1|X_2,X_3} = \frac{1/8}{1/4} = \frac{1}{2}$$

(e) Conditional

$$f_{X_1,X_2|X_3} = \frac{1/8}{1/2} = \frac{1}{4}$$

(f) Independence

Verified using LHS = RHS

Example 3: Trivariate Exponential

Given:

$$f_{XYZ}(x,y,z) = Ke^{-(ax+by+cz)}, \quad x,y,z > 0$$

(a) Constant

$$\int_0^\infty \int_0^\infty \int_0^\infty Ke^{-(ax+by+cz)} dxdydz = 1$$

Factorization:

$$K\left(\frac{1}{a}\right)\left(\frac{1}{b}\right)\left(\frac{1}{c}\right) = 1 \Rightarrow K = abc$$

(b) $f_{XY}(x,y)$

$$\int_0^\infty abce^{-(ax+by+cz)} dz$$
$$= abe^{-(ax+by)}$$

(c) $f_X(x)$

$$\int_0^\infty abe^{-(ax+by)} dy$$
$$= ae^{-ax}$$

(d) Independence

Verified using factorization

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6. Key Results Summary

- Joint Gaussian PDF formula with correlation
- $\rho = \frac{\text{cov}}{\sigma_X \sigma_Y}$
- Marginals of jointly Gaussian are Gaussian
- Standardization: $Z = \frac{X-\mu}{\sigma}$
- $P(X < 0) = Q(2)$
- Joint PMF, CDF, PDF definitions for N variables
- Independence ⇔ product of marginals
- Uniform 3D: $C = 1/8$, marginals constant
- Exponential joint: $K = abc$, independence holds

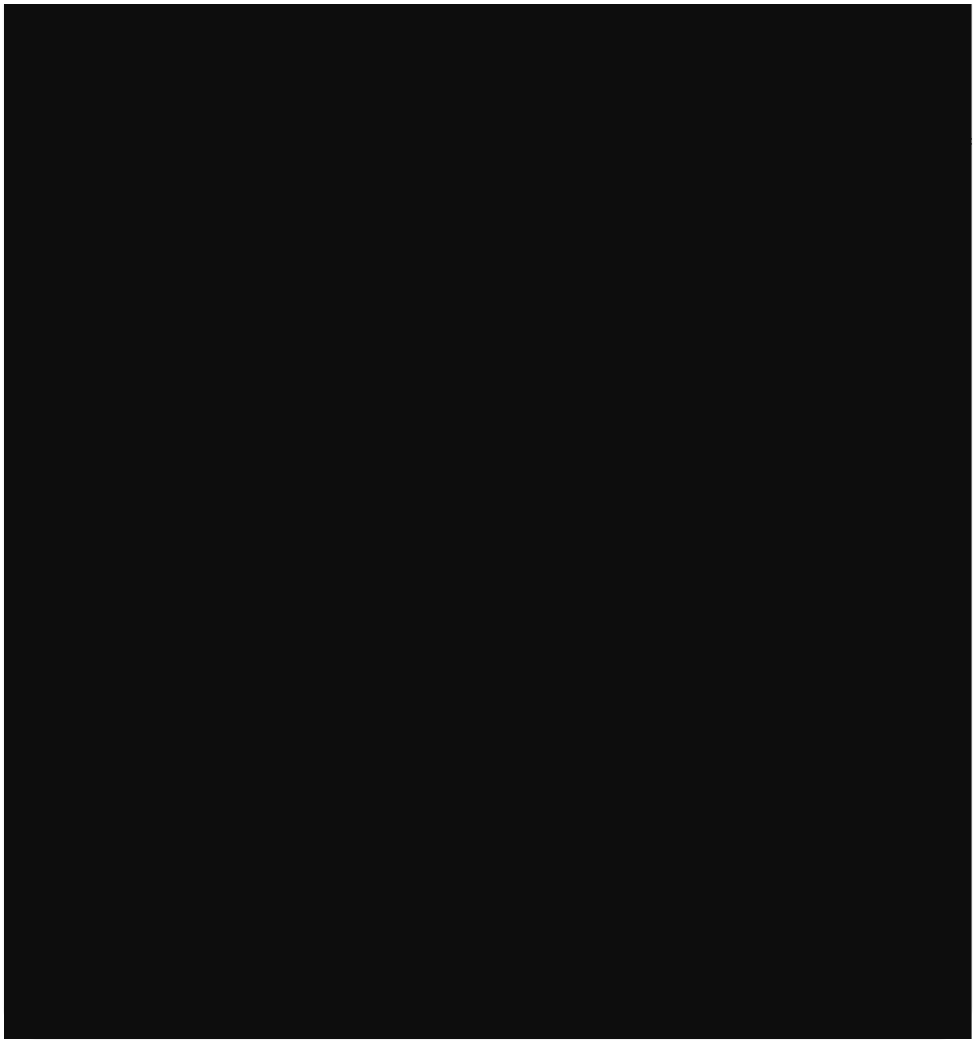
If you want, I can convert this into a **1-page cheat sheet** or **exam revision flash notes**.

convert this into latex code

Here is your **lecture scribe converted into clean LaTeX code** (exam-ready, structured exactly as requested):

LaTeX

[redacted]



If you want next level:

- I can **compress this into 1-page LaTeX cheat sheet**
- or **add proper spacing + exam formatting (boxes, highlights)**